

Name: _____

Date: _____

STRAIGHT-LINE GRAPHS, (GRADIENT AND Y-INTERCEPT)

LO: I can find the gradient and y-intercept of a line from its graph or equation.

Gradient and y-intercept

The **gradient** measures how steep a line is. Positive gradient goes up from left to right; negative goes down. The **y-intercept** is where the line crosses the y-axis ($x = 0$).

Reading from a graph: Pick two points, count the rise (up/down) and run (across). Gradient = $\text{rise} \div \text{run}$.

Quick examples

●	Rise 6, run 2 gradient = 3	divide rise by run
●	Line goes down left to right negative gradient	sign matters
●	Crosses y-axis at (0, -4) $c = -4$	read from graph
●	Rise 2, run 6 gradient = $\frac{2}{6}$	$\frac{2}{6} = \frac{1}{3}$

Key idea

Gradient = $\text{rise} \div \text{run}$ (vertical change $\div$ horizontal change).
Read the y-intercept where $x = 0$.

$$\text{gradient} = \text{rise} \div \text{run}$$

Positive = uphill, negative = downhill (left to right)

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - positive gradient from graph

A line passes through (1, 2) and (4, 8).
Find the gradient.

$$\text{Rise} = 8 - 2 = 6$$

$$\text{Run} = 4 - 1 = 3$$

$$\text{Gradient} = 6 \div 3 = 2$$

$$\text{Gradient} = 2$$

Count squares up (rise) and across (run) between two clear points.

Example 2 - negative gradient

A line passes through (0, 6) and (3, 0).
Find the gradient.

$$\text{Rise} = 0 - 6 = -6$$

$$\text{Run} = 3 - 0 = 3$$

$$\text{Gradient} = -6 \div 3 = -2$$

$$\text{Gradient} = -2$$

The line goes down, so the gradient is negative.

Example 3 - y-intercept from graph

A line crosses the y-axis at (0, -3).
What is c ?

The y-intercept is where $x = 0$

The line crosses at $y = -3$

$$\text{So } c = -3$$

$$c = -3$$

Read the y-coordinate where the line meets the y-axis.

Example 4 - from equation

Find the gradient and y-intercept of $2y = 6x - 8$.

$$\text{Divide by 2: } y = 3x - 4$$

Compare with $y = mx + c$

Gradient = 3, y-intercept = -4

$$\text{Gradient} = 3, \text{ y-intercept} = -4$$

Rearrange to $y = mx + c$ form to read off m and c .

YOUR TURN – PRACTICE (deliberate practice)

1. Gradient from two points

Find the gradient of the line through each pair.

- a) (0, 1) and (3, 7)
- b) (1, 5) and (4, 11)
- c) (0, 8) and (4, 0)
- d) (2, 3) and (5, 12)

2. Identifying positive and negative

State whether the gradient is positive, negative, or zero.

- a) Line goes uphill left to right
- b) Line goes downhill left to right
- c) Horizontal line
- d) $y = 5 - 3x$

3. y-intercept

Find the y-intercept of each line.

- a) $y = 4x + 7$
- b) $y = -x - 2$
- c) $y = 3x$
- d) $2y = 4x + 6$

4. Complete the equation

Find the equation $y = mx + c$ from the gradient and y-intercept.

- a) Gradient 4, y-intercept 3
- b) Gradient -2, y-intercept 5
- c) Gradient $\frac{1}{2}$, y-intercept -1
- d) Gradient -3, y-intercept 0

5. Mixed gradient problems

Solve each problem.

- a) Points (0, 2) and (5, 12). Find gradient and equation.
- b) A line has gradient 3 and passes through (2, 8). Find c.
- c) Which is steeper: $y = 5x + 1$ or $y = 3x - 4$?
- d) Line through (-1, 7) and (1, 3). Gradient?
- e) A horizontal line through (3, 5). Equation?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Gradient = rise ÷ run ☐

Correct answer: _____

Reason: _____

b) A line going downhill has positive gradient ☐

Correct answer: _____

Reason: _____

c) The y-intercept is where the line crosses the y-axis ☐

Correct answer: _____

Reason: _____

d) A steeper line always has a smaller gradient ☐

Correct answer: _____

Reason: _____

e) A horizontal line has gradient 0 ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A line passes through (3, 7) and has gradient -2. Find its equation. At what point does it cross the x-axis? What is the area of the triangle formed by this line and the two axes?

Expression: _____

Simplified: _____

b) Three points are given: A(-1, 1), B(2, 7), C(5, 13). Show that all three points lie on the same straight line (are collinear). Find the equation of this line.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

